

NETWORK SECURITY (CCF2C01)

AY2025/2026 OCT SEMESTER

Project Report Submission

Practical Class: P01

Group Number: 2

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Submission Date: {02/01/2026}

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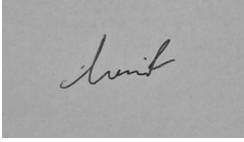
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Suggested Report Format / Contents:

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1. Introduction

This project documents the design and implementation of a secure, decentralized enterprise network for Group 2, which consists of three primary locations: Tampines, Sengkang, and Bedok. The central focus of the project was to unify these sites into a single, cohesive communications mesh using the public internet as the transport medium. Because each site acts as an equal peer in the network, we utilized a symmetrical architecture across all three locations. This approach ensures that security policies, routing protocols, and management procedures are consistent, which is essential for a stable and predictable production environment.

The Proof of Concept (PoC) validates the integration of several critical security and networking technologies. We focused on the implementation of stateful firewalling to manage traffic flow, encrypted VPN tunneling to protect data in transit, and efficient address management using Variable Length Subnet Masking (VLSM). By moving away from a traditional hub-and-spoke model and treating every location as an equal peer, we ensured that inter-site communication is direct and does not rely on a central aggregation point, which optimizes routing and reduces latency for cross-site traffic.

In addition to the network implementation, this project also explores the concept of Zero Trust as part of our security research. Zero Trust is a modern security model that challenges the traditional assumption of implicit trust within internal networks, instead emphasizing strict verification, segmentation, and least-privilege access. While a full Zero Trust architecture is not implemented in this proof-of-concept, the principles were studied and analysed to understand how enterprise networks can further strengthen their security posture beyond perimeter-based defenses. This research provides context for evaluating potential future improvements to the network design.

1.1 Overview

The project is built around three sites that share a standardized internal design, which was necessary to streamline the integration process between the team members. The primary goal was to merge these sites securely over the untrusted public internet using industry-standard protocols.

To achieve this, we first established a Site-to-Site VPN mesh. This involves the use of IPsec (Internet Protocol Security) to create encrypted tunnels between the sites. These tunnels encapsulate private IP packets within encrypted headers, ensuring that corporate data remains confidential and protected from interception while traveling across the ISP's infrastructure. Additionally, because the organization operates with a limited public IP allocation for Group 2, we implemented Network Address Translation (NAT) specifically at the exit router of each site. This allows internal hosts within the private LAN and DMZ segments to access external internet services by sharing a pool of public IP addresses assigned by the ISP.

At the edge of each site, we deployed a Cisco ASA 5506-X firewall. The ASA functions as the primary security gatekeeper by performing stateful packet inspection on all inbound and outbound traffic. This means the firewall tracks the state of active connections and only allows return traffic that matches an existing internal request. We further refined the security posture by implementing hierarchical security zones. By assigning specific security levels—INSIDE (100),

DMZ (50), and OUTSIDE (1)—we established clear trust boundaries. These levels dictate that traffic flowing from a high-security zone to a lower-security zone is permitted by default, while traffic moving in the opposite direction is strictly denied unless explicitly permitted by an Access Control List (ACL).

1.2 Objectives:

The primary technical objectives of the project were centered on the secure integration of resources and the systematic hardening of the network infrastructure. Our first priority was the establishment of the VPN mesh to facilitate secure data transfers between Tampines, Sengkang, and Bedok. This required precise coordination of encryption algorithms and hashing protocols between the three sites to ensure that the tunnels could be successfully negotiated and maintained.

Another core objective was the implementation of NAT to provide internet access for each site's private segments. By configuring NAT on the exit routers rather than the firewalls, we adhered to the specific design requirements of the project while ensuring that internal private IP addressing remained hidden from the public internet. We also utilized the Cisco ASA to protect critical network resources. This involved creating granular access rules to permit only necessary services, such as HTTP/HTTPS for web servers or NTP for clock synchronization, while blocking all other unauthorized ports.

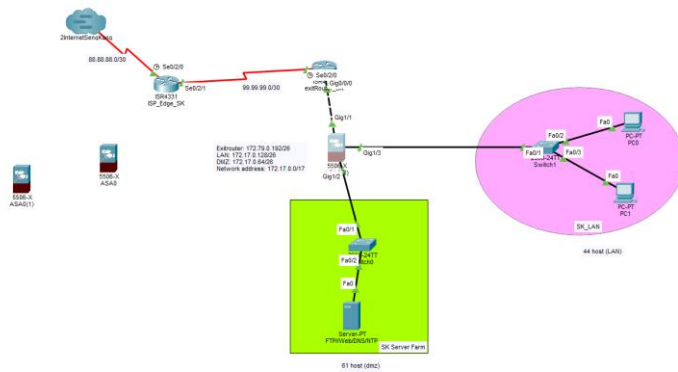
Finally, we focused on the comprehensive hardening of the management plane. This included enforcing secure remote access through SSH version 2.0 and implementing AAA (Authentication, Authorization, and Accounting) to manage administrative credentials locally. We used the netAdmin account for routers and switches and the Admin01 account for the ASA firewall as per the specifications. To ensure the integrity of our forensic logs, we synchronized all devices as NTP clients to a central DMZ server. This ensures that every security event recorded across the three sites shares a common, accurate timestamp, which is vital for correlating data during an incident response or a security audit.

Part A – Packet Tracer Configuration

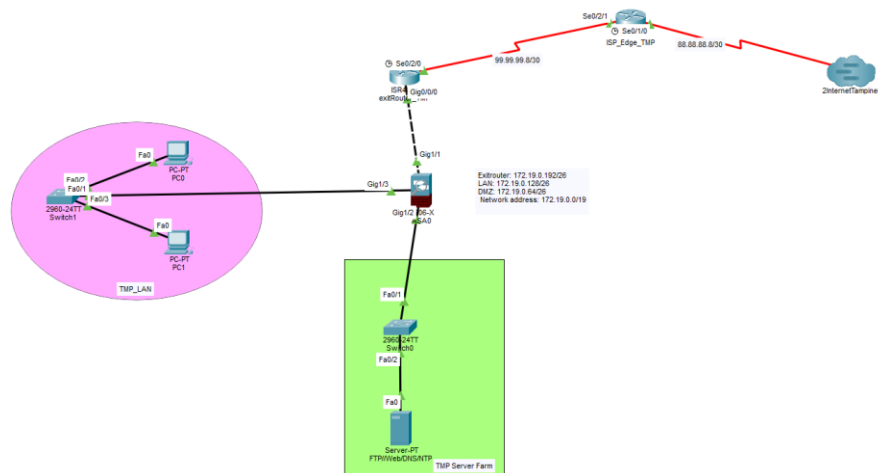
2. Network Design

Describe the overall network topology including the core internet cloud and individual sites. Include diagrams to illustrate the design.

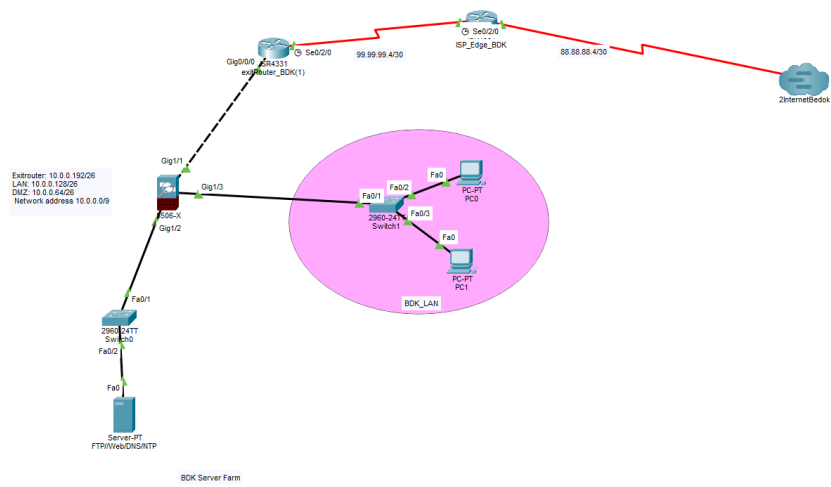
Sengkang: (Done by Lucas Chong)



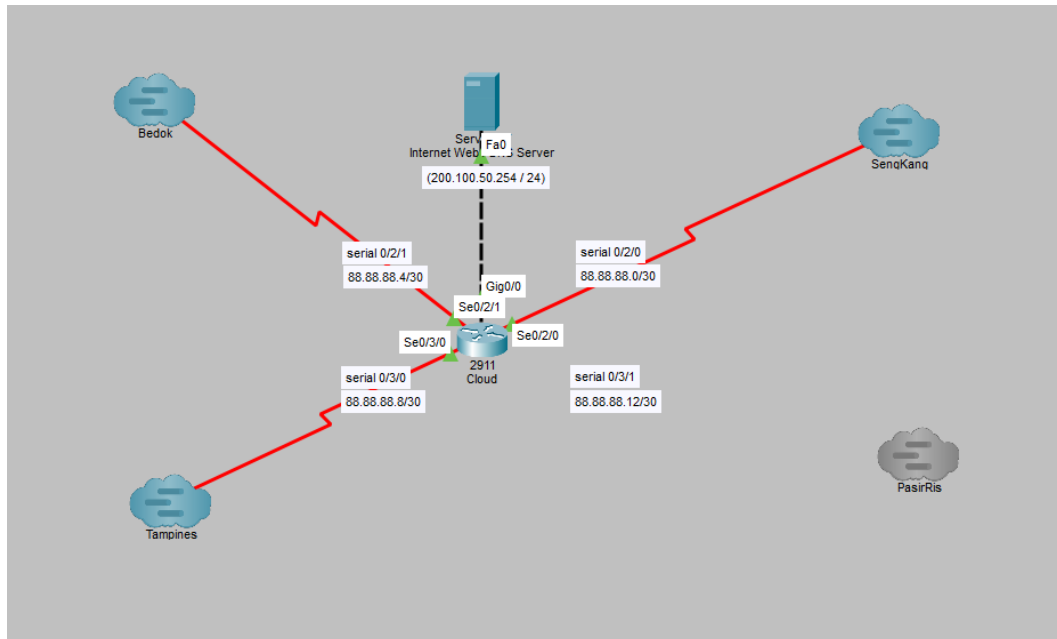
Tampines: (Done by Hasif)



Bedok: (Done by Lucas Joshua)



ISP Core Internet Cloud



3. Configuration Details

3.1 IP Addressing Scheme

Describe the IP addressing scheme used for each site based on the group assignment.

Sengkang Site

Network segment	Host Needed	Host Available	Subnet Address	CIDR	Usable Range	Gateway
LAN	61	62	255.255.255.192	/26	172.17.0.65-172.17.0.126	172.17.0.65
DMZ Server Farm	61	62	255.255.255.192	/26	172.17.0.129-172.17.0.190	172.17.0.129
To ExitRouter	61	62	255.255.255.192	/26	172.17.0.193-172.17.0.194	172.17.0.194

Tampines Site

Network segment	Host Needed	Host Available	Subnet Address	CIDR	Usable Range	Gateway
DMZ Server Farm Hosts	35	62	255.255.255.192	/26	172.19.0.65-172.19.0.126	172.19.0.65
LAN	35	62	255.255.255.192	/26	172.19.0.129-172.19.0.190	172.19.0.129
ASA-FW to ExitRouter	35	62	255.255.255.192	/26	172.19.0.193-172.19.0.254	172.19.0.194

Bedok Site

Network segment	Host Needed	Host Available	Subnet Address	CIDR	Usable Range	Gateway
DMZ Server Farm Hosts	30	62	255.255.255.192	/26	10.0.0.65 - 10.0.0.126	10.0.0.65
LAN	30	62	255.255.255.192	/26	10.0.0.129 - 10.0.0.190	10.0.0.129
ASA-FW to ExitRouter	30	62	255.255.255.192	/26	10.0.0.193 - 10.0.0.254	10.0.0.193

Although the required host is 30 and a CIDR of /27 will suffice, Bedok is set as /26 so that host available would be 62. This allows for **scalability** in the future.

3.2 VPN Configuration

VPN was configured between BEDOK, SENGKANG, and TAMPINES

The configurations are run through below, and testing will be done in 4.1

3.21 VPN: Security parameters:

VPN Security parameters are the technical standards and configurations used to establish a secure, encrypted tunnel for data transmission. They include encryption, secure protocols (IPsec), authentication methods, and other ways to ensure data integrity and confidentiality

The security parameters can be shown in cisco packet tracer by running:

Show crypto isakmp policy

```
Global IKE policy
Protection suite of priority 10
  encryption algorithm:    AES - Advanced Encryption Standard (256 bit keys)
  hash algorithm:          Secure Hash Standard
  authentication method:   Pre-Shared Key
  Diffie-Hellman group:    #2 (1024 bit)
  lifetime:                 86400 seconds, no volume limit
```

- This is set for all 3 sites

Justification of configurations

Protection suite of priority 10	This is a VPN policy 'ID' with priority 10, if no lower priority lists, offers this specific ruleset to its peers during the handshake
encryption algorithm: AES - Advanced Encryption Standard (256 bit keys).	The encryption method for data encryption 256 bits is the industry standard, as the default 128 is much easier to brute force
hash algorithm: Secure Hash Standard	Ensures data integrity remains, data unchanged. If the codes match, the receiver will know the packet was not altered
authentication method: Pre-Shared Key	Both routers are configured with the same secret string "iLOVEnws". If the keys do not match, the routers do not form peers
Diffie-Hellman group: #2 (1024 bit)	Allows peer routers to create shared secret key over a insecure public network. Group 2: uses 1024-bit prime numbers, making it complex and difficult to crack the secret key
lifetime: 86400 seconds, no volume limit	Exactly every 24 hours, the router will tear down the management tunnel and negotiate brand-new keys, ensuring hackers cannot reuse old keys

3.22 VPN: Transform set configuration

We are using tunnel mode, which encapsulates and encrypts the entire original IP packet within a new separate IP packet, primary used for Site-to-Site vpn (what we are doing)

The transform set is created by doing:

```
crypto ipsec transform-set TS-TRANS esp-aes 256 esp-sha-hmac
```

esp-aes 256 esp-sha-hmac -> syntax explained in the table below

Command: show crypto ipsec transform-set TS-TRANS

On Tampines, bedok, and Sengkang

```
Transform set TS-TRANS: {      { esp-256-aes esp-sha-hmac  }
    will negotiate = { Tunnel,  },

Transform set $!default_transform_set_1: { esp-aes esp-sha-hmac  }
    will negotiate = { Transport,  },
Transform set $!default_transform_set_0: { esp-3des esp-sha-hmac  }
    will negotiate = { Transport,  },
```

Explanation of output:

Transform set TS-TRANS	The transform-set identified for the Phase 2 security policy
esp-256-aes	256-bit AES encryption, providing high confidentiality for data
esp-sha-hmac	SHA hashing, providing data integrity and origin authentication

Tunnel mode	The mode used for TS-TRANS, required for Site-to-Site VPNs to hide private IP headers
Transform Mode	Default Cisco mode but not in use (using TS-TRANS)

3.23 VPN: Application and Traffic

To establish proper Site-To-Site VPN, other than matching security parameters, **interesting traffic ACL** has to mirror each other, mirroring **preshaed crypto keys** has to be generated with **matching passwords**.

Interesting traffic ACL:

Bedok

```
exitRouterBK#show access-list
Extended IP access list VPN-SENGKANG
 10 permit ip 10.0.0.64 0.0.0.63 172.17.0.64 0.0.0.63
 20 permit ip 10.0.0.64 0.0.0.63 172.17.0.128 0.0.0.63
 30 permit ip 10.0.0.128 0.0.0.63 172.17.0.64 0.0.0.63
 40 permit ip 10.0.0.128 0.0.0.63 172.17.0.128 0.0.0.63
Extended IP access list VPN-TAMPINES
 10 permit ip 10.0.0.64 0.0.0.63 172.19.0.64 0.0.0.63
 20 permit ip 10.0.0.64 0.0.0.63 172.19.0.128 0.0.0.63
 30 permit ip 10.0.0.128 0.0.0.63 172.19.0.64 0.0.0.63
 40 permit ip 10.0.0.128 0.0.0.63 172.19.0.128 0.0.0.63
```

Tampines

```
ExitRouter_SK#show access-lists
Extended IP access list VPN-BEDOK
 10 permit ip 172.17.0.64 0.0.0.63 10.0.0.64 0.0.0.63
 20 permit ip 172.17.0.64 0.0.0.63 10.0.0.128 0.0.0.63
 30 permit ip 172.17.0.128 0.0.0.63 10.0.0.64 0.0.0.63
 40 permit ip 172.17.0.128 0.0.0.63 10.0.0.128 0.0.0.63
Extended IP access list VPN-TAMPINES
 10 permit ip 172.17.0.64 0.0.0.63 172.19.0.64 0.0.0.63
 20 permit ip 172.17.0.64 0.0.0.63 172.19.0.128 0.0.0.63
 30 permit ip 172.17.0.128 0.0.0.63 172.19.0.64 0.0.0.63
 40 permit ip 172.17.0.128 0.0.0.63 172.19.0.128 0.0.0.63
```

Sengkang

```
exitRouter_TMP#show access-lists
Extended IP access list VPN-BEDOK
 10 permit ip 172.19.0.64 0.0.0.63 10.0.0.64 0.0.0.63
 20 permit ip 172.19.0.64 0.0.0.63 10.0.0.128 0.0.0.63
 30 permit ip 172.19.0.128 0.0.0.63 10.0.0.64 0.0.0.63
 40 permit ip 172.19.0.128 0.0.0.63 10.0.0.128 0.0.0.63
Extended IP access list VPN-SENGKANG
 10 permit ip 172.19.0.64 0.0.0.63 172.17.0.64 0.0.0.63
 20 permit ip 172.19.0.64 0.0.0.63 172.17.0.128 0.0.0.63
 30 permit ip 172.19.0.128 0.0.0.63 172.17.0.64 0.0.0.63
 40 permit ip 172.19.0.128 0.0.0.63 172.17.0.128 0.0.0.63
```

These access-lists were configured to identify interesting traffic as Private IPs from a site to the two peer subnets. They have to be mirrored or the Site-To-Site connection cannot be established.

Preshared Crypto Keys with matching passwords

Bedok

```
crypto isakmp key iLOVEnews address 99.99.99.1
crypto isakmp key iLOVEnews address 99.99.99.9
```

ExitRouterIP of Bedok: 99.99.99.5

Tampines

```
crypto isakmp key iLOVEnews address 99.99.99.1
crypto isakmp key iLOVEnews address 99.99.99.5
```

ExitRouterIP of Tampines: 99.99.99.9

Sengkang

```
crypto isakmp key iLOVEnews address 99.99.99.5
crypto isakmp key iLOVEnews address 99.99.99.9
```

ExitRouterIP of Sengkang: 99.99.99.1

Each of these crypto isakmp key passwords is iLOVEnews, used to authenticated it peers

3.24 VPN: Crypto map

Crypto map (commands ran)

Bedok

```
exitRouterBK#show run | section crypto map
crypto map VPN-MAP 10 ipsec-isakmp
 set peer 99.99.99.1
 set pfs group2
 set security-association lifetime seconds 1800
 set transform-set TS-TRANS
 match address VPN-SENGKANG
crypto map VPN-MAP 20 ipsec-isakmp
 set peer 99.99.99.9
 set pfs group2
 set security-association lifetime seconds 1800
 set transform-set TS-TRANS
 match address VPN-TAMPINES
crypto map VPN-MAP
```

Tampines

```
ExitRouter_SK#show run | section crypto map
crypto map VPN-MAP 10 ipsec-isakmp
  set peer 99.99.99.5
  set pfs group2
  set security-association lifetime seconds 1800
  set transform-set TS-TRANS
  match address VPN-BEDOK
crypto map VPN-MAP 20 ipsec-isakmp
  set peer 99.99.99.9
  set pfs group2
  set security-association lifetime seconds 1800
  set transform-set TS-TRANS
  match address VPN-TAMPINES
crypto map VPN-MAP
```

Sengkang

```
exitRouter_TMP#show run | section crypto map
crypto map VPN-MAP 10 ipsec-isakmp
  set peer 99.99.99.5
  set pfs group2
  set security-association lifetime seconds 1800
  set transform-set TS-TRANS
  match address VPN-BEDOK
crypto map VPN-MAP 20 ipsec-isakmp
  set peer 99.99.99.1
  set pfs group2
  set security-association lifetime seconds 1800
  set transform-set TS-TRANS
  match address VPN-SENGKANG
crypto map VPN-MAP
```

Syntax for crypto map VPN-MAP X ipsec-isakmp

set peer <exitrouter ip>	Destination public IP address of the exitrouter. Set one per VPN-MAP, towards one Site
set pfs group2	Perfect Forward Secrecy ensures that the keys use for data encryption are not derived from the Phase 1 Keys. By using group 2, a new independent handshake is ensured every session refresh
set security-association lifetime seconds 1800	Tells the router to generate new encryption keys every 1800 seconds
set transform-set TS-TRANS	Links the tunnel mode encryption settings (AES-256/SHA) to this specific peer
match address <Accesslist name>	Points to the interesting traffic ACL

Crypto map (configurations and settings)

Bedok

```
exitRouterBK#show crypto map
Crypto Map VPN-MAP 10 ipsec-isakmp
  Peer = 99.99.99.1
  Extended IP access list VPN-SENGKANG
    access-list VPN-SENGKANG permit ip 10.0.0.64 0.0.0.63 172.17.0.64 0.0.0.63
    access-list VPN-SENGKANG permit ip 10.0.0.64 0.0.0.63 172.17.0.128 0.0.0.63
    access-list VPN-SENGKANG permit ip 10.0.0.128 0.0.0.63 172.17.0.64 0.0.0.63
    access-list VPN-SENGKANG permit ip 10.0.0.128 0.0.0.63 172.17.0.128 0.0.0.63
  Current peer: 99.99.99.1
  Security association lifetime: 4608000 kilobytes/1800 seconds
  PFS (Y/N): Y
  Transform sets={
    TS-TRANS,
  }
  Interfaces using crypto map VPN-MAP:
    Serial0/2/0

Crypto Map VPN-MAP 20 ipsec-isakmp
  Peer = 99.99.99.9
  Extended IP access list VPN-TAMPINES
    access-list VPN-TAMPINES permit ip 10.0.0.64 0.0.0.63 172.19.0.64 0.0.0.63
    access-list VPN-TAMPINES permit ip 10.0.0.64 0.0.0.63 172.19.0.128 0.0.0.63
    access-list VPN-TAMPINES permit ip 10.0.0.128 0.0.0.63 172.19.0.64 0.0.0.63
    access-list VPN-TAMPINES permit ip 10.0.0.128 0.0.0.63 172.19.0.128 0.0.0.63
  Current peer: 99.99.99.9
  Security association lifetime: 4608000 kilobytes/1800 seconds
  PFS (Y/N): Y
  Transform sets={
    TS-TRANS,
  }
  Interfaces using crypto map VPN-MAP:
    Serial0/2/0
```

Tampines

```
Crypto Map VPN-MAP 10 ipsec-isakmp
  Peer = 99.99.99.5
  Extended IP access list VPN-BEDOK
    access-list VPN-BEDOK permit ip 172.19.0.64 0.0.0.63 10.0.0.64 0.0.0.63
    access-list VPN-BEDOK permit ip 172.19.0.64 0.0.0.63 10.0.0.128 0.0.0.63
    access-list VPN-BEDOK permit ip 172.19.0.128 0.0.0.63 10.0.0.64 0.0.0.63
    access-list VPN-BEDOK permit ip 172.19.0.128 0.0.0.63 10.0.0.128 0.0.0.63
  Current peer: 99.99.99.5
  Security association lifetime: 4608000 kilobytes/1800 seconds
  PFS (Y/N): Y
  Transform sets={
    TS-TRANS,
  }
  Interfaces using crypto map VPN-MAP:
    Serial0/2/0

Crypto Map VPN-MAP 20 ipsec-isakmp
  Peer = 99.99.99.1
  Extended IP access list VPN-SENGKANG
    access-list VPN-SENGKANG permit ip 172.19.0.64 0.0.0.63 172.17.0.64 0.0.0.63
    access-list VPN-SENGKANG permit ip 172.19.0.64 0.0.0.63 172.17.0.128 0.0.0.63
    access-list VPN-SENGKANG permit ip 172.19.0.128 0.0.0.63 172.17.0.64 0.0.0.63
    access-list VPN-SENGKANG permit ip 172.19.0.128 0.0.0.63 172.17.0.128
0.0.0.63
  Current peer: 99.99.99.1
  Security association lifetime: 4608000 kilobytes/1800 seconds
  PFS (Y/N): Y
  Transform sets={
    TS-TRANS,
  }
  Interfaces using crypto map VPN-MAP:
    Serial0/2/0
```

Sengkang

```
Crypto Map VPN-MAP 10 ipsec-isakmp
Peer = 99.99.99.5
Extended IP access list VPN-BEDOK
  access-list VPN-BEDOK permit ip 172.17.0.64 0.0.0.63 10.0.0.64 0.0.0.63
  access-list VPN-BEDOK permit ip 172.17.0.64 0.0.0.63 10.0.0.128 0.0.0.63
  access-list VPN-BEDOK permit ip 172.17.0.128 0.0.0.63 10.0.0.64 0.0.0.63
  access-list VPN-BEDOK permit ip 172.17.0.128 0.0.0.63 10.0.0.128 0.0.0.63
Current peer: 99.99.99.5
Security association lifetime: 4608000 kilobytes/1800 seconds
PFS (Y/N): Y
Transform sets={
  TS-TRANS,
}
Interfaces using crypto map VPN-MAP:
  Serial0/2/0

Crypto Map VPN-MAP 20 ipsec-isakmp
Peer = 99.99.99.9
Extended IP access list VPN-TAMPINES
  access-list VPN-TAMPINES permit ip 172.17.0.64 0.0.0.63 172.19.0.64 0.0.0.63
  access-list VPN-TAMPINES permit ip 172.17.0.64 0.0.0.63 172.19.0.128 0.0.0.63
  access-list VPN-TAMPINES permit ip 172.17.0.128 0.0.0.63 172.19.0.64 0.0.0.63
  access-list VPN-TAMPINES permit ip 172.17.0.128 0.0.0.63 172.19.0.128
0.0.0.63
Current peer: 99.99.99.9
Security association lifetime: 4608000 kilobytes/1800 seconds
PFS (Y/N): Y
Transform sets={
  TS-TRANS,
}
Interfaces using crypto map VPN-MAP:
  Serial0/2/0
```

These commands show the correct application of **access-list** applied in each cryptomap and each peer, as well as the correct **interface** using cryptomap. (Serial0/2/0)

Summary

With these matching/mirroring configurations, the VPN tunnels theoretically should be able to connect and be established Site-To-Site

3.3 NAT Configuration

At each site, NAT is configured on the site firewall (ASA) positioned between internal networks (LAN and DMZ) and the exit router that connects to the internet.

Each Site uses a dynamic NAT with a public address pool, allowing multiple internal hosts to access internet services using a limited range of public IP addresses. The public IP pool assigned to each site is provided by the ISP and is unique per site to avoid address conflicts.

The NAT configuration uses a standard zone firewall design:

- INSIDE: Internal user LAN
- DMZ: Server farm hosting internal services
- OUTSIDE: Connection to the exit router and internet

NAT is applied when traffic flows from higher-security interfaces (inside or DMZ) to the lower-security outside interface

The pool is configured on the ASA and is used to dynamically translate internal private addresses when accessing the Internet. Using a pool rather than a single IP improves scalability and reflects a realistic enterprise NAT deployment.

NAT is also tightly integrated with the ASA security policy. Only traffic explicitly permitted by ACLs is translated and forwarded, ensuring:

- Internal hosts cannot access the internet unless allowed by policy
- Unsolicited inbound traffic from the Internet is blocked by default
- NAT does not override firewall security rules

As a result, NAT provides address translation but does not weaken the security posture of the site network

Bedok Public IPs

66.220.149.0/25

Bedok NAT Pool and interfaces

```
ip nat pool SITE-NAT 154.26.131.1 154.26.131.254 netmask 255.255.255.0

interface GigabitEthernet0/0/0
ip address 10.0.0.194 255.255.255.192
ip nat inside
duplex auto
speed auto

interface Serial0/2/0
ip address 99.99.99.5 255.255.255.252
ip nat outside
clock rate 2000000
crypto map VPN-MAP
```

Tampines Public IPs

154.26.131.0/24

Tampines NAT Pool and interfaces

```
ip nat pool TMP-NAT-POOL 66.220.149.1 66.220.149.126 netmask 255.255.255.128

interface Serial0/2/0
description Connection to ISP_Edge_TP
ip address 99.99.99.9 255.255.255.252
ip nat outside
clock rate 2000000
crypto map VPN-MAP
!

interface GigabitEthernet0/0/0
description Connection to ASA OUTSIDE
ip address 172.19.0.194 255.255.255.192
ip nat inside
duplex auto
speed auto
```

Sengkang Public IPs

64.233.185.112/28

Sengkang NAT Pool and interfaces

```
ip nat pool SK-PUBLIC 64.233.185.113 64.233.185.126 netmask 255.255.255.240

interface Serial0/2/0
ip address 99.99.99.1 255.255.255.252
ip nat outside
clock rate 2000000
crypto map VPN-MAP
!

interface GigabitEthernet0/0/0
ip address 172.17.0.194 255.255.255.192
ip nat inside
duplex auto
speed auto
```

The access-list for NAT translations will be shown in 3.31

The proof of IP NAT Translations will be shown in 4.2 (internet access)

3.31 NAT vs VPN Conflicts (NAT EXEMPT)

NAT and VPN are both configured at the exit router. However, the interesting traffic for VPN conflicts with the translating of private addresses in NAT. As a result, an **extended ACL** has to be added. This is known as **NAT Exemption**. The extended ACL will:

- Deny traffic specifically to subnet of other sites
- Allow destination traffic to anywhere else (This is the traffic that will undergo NAT translations)
- The denied traffic (Interesting traffic for VPN) from the access-list will bypass the NAT process, and instead be picked up by crypto map and undergo encapsulation and encryption instead

NAT Exempt for each site + NAT overload source list

Bedok

```
Extended IP access list NAT-EXEMPT-BDK
10 deny ip 10.0.0.64 0.0.0.63 172.17.0.64 0.0.0.63
20 deny ip 10.0.0.64 0.0.0.63 172.17.0.128 0.0.0.63
30 deny ip 10.0.0.64 0.0.0.63 172.19.0.64 0.0.0.63
40 deny ip 10.0.0.64 0.0.0.63 172.19.0.128 0.0.0.63
50 deny ip 10.0.0.128 0.0.0.63 172.17.0.64 0.0.0.63
60 deny ip 10.0.0.128 0.0.0.63 172.17.0.128 0.0.0.63
70 deny ip 10.0.0.128 0.0.0.63 172.19.0.64 0.0.0.63
80 deny ip 10.0.0.128 0.0.0.63 172.19.0.128 0.0.0.63
90 permit ip 10.0.0.64 0.0.0.63 any
100 permit ip 10.0.0.128 0.0.0.63 any (4 match(es))
```

```
ip nat inside source list NAT-EXEMPT-BDK pool SITE-NAT overload
```

Tampines

```
Extended IP access list NAT-EXEMPT-TMP
10 deny ip 172.19.0.64 0.0.0.63 172.17.0.64 0.0.0.63
20 deny ip 172.19.0.64 0.0.0.63 172.17.0.128 0.0.0.63
30 deny ip 172.19.0.128 0.0.0.63 172.17.0.64 0.0.0.63
40 deny ip 172.19.0.128 0.0.0.63 172.17.0.128 0.0.0.63
50 deny ip 172.19.0.64 0.0.0.63 10.0.0.64 0.0.0.63
60 deny ip 172.19.0.64 0.0.0.63 10.0.0.128 0.0.0.63
70 deny ip 172.19.0.128 0.0.0.63 10.0.0.64 0.0.0.63
80 deny ip 172.19.0.128 0.0.0.63 10.0.0.128 0.0.0.63
90 permit ip 172.19.0.64 0.0.0.63 any
100 permit ip 172.19.0.128 0.0.0.63 any (2 match(es))
```

```
ip nat inside source list NAT-EXEMPT-TMP pool TMP-NAT-POOL overload
```

Sengkang

```
Extended IP access list NAT-EXEMPT-SENGKANG
10 deny ip 172.17.0.64 0.0.0.63 172.19.0.128 0.0.0.63
20 deny ip 172.17.0.64 0.0.0.63 172.19.0.64 0.0.0.63
30 deny ip 172.17.0.128 0.0.0.63 172.19.0.64 0.0.0.63
40 deny ip 172.17.0.128 0.0.0.63 172.19.0.128 0.0.0.63
50 deny ip 172.17.0.64 0.0.0.63 10.0.0.64 0.0.0.63
60 deny ip 172.17.0.64 0.0.0.63 10.0.0.128 0.0.0.63
70 deny ip 172.17.0.128 0.0.0.63 10.0.0.64 0.0.0.63
80 deny ip 172.17.0.128 0.0.0.63 10.0.0.128 0.0.0.63
90 permit ip 172.17.0.64 0.0.0.63 any
100 permit ip 172.17.0.128 0.0.0.63 any
```

```
ip nat inside source list NAT-EXEMPT-SENGKANG pool SK-PUBLIC overload
```

- Each access list denies traffic from LAN and DMZ subnets in a site to LAN and DMZ one another. (Interested Traffic for VPN)
- If the traffic is permitted from the access-list, they will undergo PAT translations.
- If it is denied, it undergoes VPN Tunneling and is set via Site-To-Site

In summary, NAT is configured at each site using dynamic address translation with a public IP pool on the ASA firewall. This design allows internal LAN and DMZ hosts to access public Internet services while preserving private IP addressing and maintaining strong security controls. The NAT configuration aligns with enterprise best practices and supports scalable, secure Internet connectivity for all sites in the network.

3.4 ASA Firewall Configuration

The ASA firewall has many configurations, which will be gone through in sections. The basic overview of fundamental configurations include: (Done on all 3 Firewalls)

```
hostname Site-ASA5506
domain-name NWS.com
enable password N37XLieiShxS1BPc encrypted
```

- Changing the hostname
- Changing the domain-name to NWS.COM
- Enabling password iLOVE nws upon trying to enable the ASA firewall

3.41 ASA(DHCP)

Due to Cisco Packet Tracer limitations, Cisco Packet Tracer (CPT) version 9.0.0 cannot be used for dhcpd on the ASA. We have downgrade to use version 8.2.2 to configure and test DHCP accordingly

An example of DHCP configuration of Sengkang Site:

dhcpd address 172.17.0.139-172.17.0.190 inside – sets the range of IP addresses that the inside interface can receive

dhcpd dns 200.100.50.254 interface inside – sets the dns IP of DHCP assigned device

dhcpd enable inside – enables DHCP to the inside interface

Bedok

```
SITE-ASA5506#show run dhcpd
dhcpd address 10.0.0.139-10.0.0.190 inside
dhcpd dns 200.100.50.254 interface inside
dhcpd enable inside
!
```

Tampines

```
Site-ASA5506#show run dhcpd
dhcpd address 172.19.0.139-172.19.0.190 INSIDE
dhcpd dns 200.100.50.254 interface INSIDE
dhcpd enable INSIDE
```

Sengkang

```
Site-ASA5506#show run dhcpd
dhcpd address 172.17.0.139-172.17.0.190 inside
dhcpd dns 200.100.50.254 interface inside
dhcpd enable inside
```

As stated in the specifications, the DHCP address given to the lease is excluding the first 10 address in the subnet address, as the first 10 are for static IPs for servers and default gateways

3.42 ASA(AAA)

AAA stands for Authentication, Authorization, and Accounting

AAA determines what users can access the network and what users are able to access which services. By using a username and a password from the local database, the device is able to assign privileges to users according to what they are authorized to use.

Without AAA, users who know the secret password are able to access anything in the network device, without any segregation of privileges

Ran on each sites ASA-5506-X

```
SITE-ASA5506#show run | include aaa
aaa authentication ssh console LOCAL
```

aaa authentication ssh console LOCAL – is a command used to authenticate the credentials that are being used to log in into the ASA to use the username and password that is configured locally on it

- No need to reference external authentication servers.
- Authentication also remain even if network connectivity is lost

Without the command, the SSH attempt to log in can be done just by the username, neglecting the purpose of the user identification method

3.43 ASA(SSH)

In each site, the ASA is configured with a ssh command:

Ssh <IP of PC1> 255.255.255.255 inside

To only allow PC1 to ssh into the ASA, like a Network Engineer PC.

ssh timeout 20 – SSH timeout is also configured to be 20 minutes.

These two lines reduces the likelihood of rogue users within an organisation to attempt to SSH into the firewall through any other PCs, and reduce the chance of others from using social engineering to physically use PC1 when the Authorized User is logged in into SSH, but away from the PC

The ASA firewall login for SSH

ASA SSH Credentials:

```
username Admin01 password N37XLieiShxS1BPc encrypted
!
```

Bedok site statically allow SSH

```
SITE-ASA5506#show run | section ssh
aaa authentication ssh console LOCAL
ssh 10.0.0.141 255.255.255.255 inside
ssh timeout 20
```

Tampines site statically allow SSH

```
Site-ASA5506#show run | section ssh
aaa authentication ssh console LOCAL
ssh 172.19.0.130 255.255.255.255 INSIDE
ssh timeout 20
```

Sengkang site statically allow SSH

```
Site-ASA5506#show run | section ssh
aaa authentication ssh console LOCAL
ssh 172.17.0.140 255.255.255.255 inside
ssh timeout 20
```

3.44 ASA(NTP)

Network Time Protocol (NTP) is a networking protocol for clock-synchronizing across computer systems.

The ASA uses the DMZ Server as a reference clock for NTP, while using trusted key and password hashed with md5 for authentication. NTP authentication is needed to verify the NTP server identity in case of a malicious user attempts to pose as a NTP server and alter the timestamps of connected devices.

Commands ran include:

ntp authentication-key 1 md5 NTPpassword – defines the secret handshake, creates an authentication key with ID 1, and uses the MD5 hashing algorithm to set the password NTPpassword

ntp authenticate – This tells the system to only communicate with NTP sources that can prove their identity

ntp trusted-key 1 – Key ID 1 authorized for use, only trusted can be used to synchronize the clock

ntp server <IP of DMZ server> – This identifies the remote device that the ASA should contact to sync NTP

Configuring the corresponding settings on the DMZ Server, it allows connection when there is matching key ID and password

Bedok, Tampines, Sengkang DMZ Server configuration

The screenshot shows the NTP configuration interface. At the top, 'NTP' is displayed. Below it, 'Service' is set to 'On' (radio button selected). Under the 'Authentication' section, 'Enable' is selected (radio button). The 'Key' field contains the value '1', and the 'Password' field contains the value 'NTPpassword'.

(Below are the displayed configurations)

Bedok

```
SITE-ASA5506#show run | section ntp
ntp authentication-key 1 md5 *****
ntp authenticate
ntp trusted-key 1
ntp server 10.0.0.66 key 1
```

NTP->Working (bedok)

```
SITE-ASA5506#show ntp status
Clock is synchronized, stratum 2, reference is 10.0.0.66
nominal freq is 250.0000 Hz, actual freq is 249.9990 Hz, precision is 2**24
reference time is ED03E6DC.000002BB (19:5:0.699 UTC Wed Feb 4 2026)
clock offset is 1.00 msec, root delay is 4.00 msec
root dispersion is 10.17 msec, peer dispersion is 0.00 msec.
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is - 0.000001193 s/s system
poll interval is 6, last update was 11 sec ago.
SITE-ASA5506#show ntp association
```

address	ref clock	st	when	poll	reach	delay	offset
disp							
*~10.0.0.66	127.127.1.1	1	18	32	1	4.00	1.00
0.00							

* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured

Tampines

```
Site-ASA5506#show run | section ntp
ntp authentication-key 1 md5 *****
ntp authenticate
ntp trusted-key 1
ntp server 172.19.0.75
```

NTP->Working (Tampines)

```
Site-ASA5506#show ntp status
Clock is synchronized, stratum 2, reference is 172.19.0.75
nominal freq is 250.0000 Hz, actual freq is 249.9990 Hz, precision is 2**24
reference time is EC8599E2.00000164 (23:50:58.356 UTC Mon Nov 3 2025)
clock offset is 0.00 msec, root delay is 4.00 msec
root dispersion is 294.88 msec, peer dispersion is 0.48 msec.
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is - 0.000001193 s/s system
poll interval is 4, last update was 1 sec ago.
Site-ASA5506#show ntp association
```

address	ref clock	st	when	poll	reach	delay	offset
disp							
*~172.19.0.75	127.127.1.1	1	0	64	3	4.00	0.00
0.48							

* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured

```
Site-ASA5506#
```

Sengkang

```
Site-ASA5506#show run | section ntp
ntp authentication-key 1 md5 *****
ntp authenticate
ntp trusted-key 1
ntp server 172.17.0.70
```

NTP->Working (Sengkang)

```

Site-ASA5506#show ntp status
Clock is synchronized, stratum 2, reference is 172.17.0.70
nominal freq is 250.0000 Hz, actual freq is 249.9990 Hz, precision is 2**24
reference time is EC8C5AAB.000001B2 (2:47:7.434 UTC Sun Nov 9 2025)
clock offset is -1.00 msec, root delay is 4.00 msec
root dispersion is 310.47 msec, peer dispersion is 0.00 msec.
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is - 0.000001193 s/s system
poll interval is 4, last update was 1 sec ago.
Site-ASA5506#show ntp associations

address          ref clock      st  when    poll   reach  delay      offset
disp
*~172.17.0.70    127.127.1.1    1   16      16     2      4.00      -1.00
0.00
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
Site-ASA5506#

```

All 3 sites NTP clocks are **synchronised** and reference clock points to DMZ Server

3.45 ASA(Security Levels)

Traffic is allowed by default from higher security to lower security

Traffic is denied by default from lower security to higher security

Security level set for interfaces:

- INSIDE: 100
 - Reasoning: These are the company's users
 - The devices are managed internally and assumed to be safe
 - This prevents the users from internet attacks and compromised servers in DMZ
- DMZ: 50
 - Servers must be reachable, but they are also high risk targets
 - If they are compromised, they must not be able to attack users
 - Security level 50 will not allow hacked servers to attack LAN users
 - Prevents lateral movement into the internal network
- OUTSIDE: 1
 - This is the lowest trust out of the 3 as it is the public internet
 - The users are not known, and we have no control over their devices
 - The default behavior of the traffic originating from this zone is blocked
 - Only return traffic or explicit ACL-allowed traffic is permitted

(Below are the displayed configurations)

Tampines

```
interface GigabitEthernet1/1
  nameif OUTSIDE
  security-level 1
  ip address 172.19.0.193 255.255.255.192
!
interface GigabitEthernet1/2
  nameif DMZ
  security-level 50
  ip address 172.19.0.65 255.255.255.192
!
interface GigabitEthernet1/3
  nameif INSIDE
  security-level 100
  ip address 172.19.0.129 255.255.255.192
.
```

Bedok

```
interface GigabitEthernet1/1
  nameif outside
  security-level 1
  ip address 10.0.0.193 255.255.255.192
!
interface GigabitEthernet1/2
  nameif dmz
  security-level 50
  ip address 10.0.0.65 255.255.255.192
!
interface GigabitEthernet1/3
  nameif inside
  security-level 100
  ip address 10.0.0.129 255.255.255.192
.
```

Sengkang

```
interface GigabitEthernet1/1
  nameif outside
  security-level 1
  ip address 172.17.0.193 255.255.255.192
!
interface GigabitEthernet1/2
  nameif dmz
  security-level 50
  ip address 172.17.0.65 255.255.255.192
!
interface GigabitEthernet1/3
  nameif inside
  security-level 100
  ip address 172.17.0.129 255.255.255.192
.
```

Having INSIDE and OUTSIDE zones are obvious, but the reason for a third security level for having a third zone is because it gives more control over servers in the DMZ, and matches industry standard, balancing security, manageability, and accessibility

The DMZ server allows itself to be protected from outside traffic and also unable to attack the LAN interface if compromised, providing layered defence and controlled access.

With security zones, ASA is enforcing trust boundaries to block packets and not randomly.

3.46 ASA(ACLs)

Access-list is needed in ASA to allow Site-To-Site pinging in VPN. Without it, even if VPN isakmp handshake has been established, the traffic would be dropped at the ASA

Bedok

```
SITE-ASA5506#show run | include OUTSIDE-IN
access-list OUTSIDE-IN extended permit ip 172.19.0.0 255.255.224.0 10.0.0.0 255.128.0.0
access-list OUTSIDE-IN extended permit ip 172.17.0.0 255.255.192.0 10.0.0.0 255.128.0.0
access-group OUTSIDE-IN in interface outside
-----
```

Tampines

```
Site-ASA5506#show run | include OUTSIDE-IN
access-list OUTSIDE-IN extended permit ip 172.17.0.0 255.255.128.0 172.19.0.0
255.255.224.0
access-list OUTSIDE-IN extended permit ip 10.0.0.0 255.128.0.0 172.19.0.0 255.255.224.0
access-group OUTSIDE-IN in interface OUTSIDE
-----
```

Sengkang

```
access-list OUTSIDE-IN extended permit ip 172.19.0.0 255.255.224.0 172.17.0.0
255.255.128.0
access-list OUTSIDE-IN extended permit ip 10.0.0.0 255.128.0.0 172.17.0.0 255.255.128.0
!
!
access-group OUTSIDE-IN in interface outside
```

These ACLs only allow traffic from the outside if they are from the BEDOK, TAMPINES, SENGKANG subnet and is headed for the private LAN or DMZ subnets. Otherwise, by default the traffic is denied

4.1 Testing and Validation(Site-To-Site ping tests)

For reference, the devices used are:

- PC1 of Bedok site (10.0.0.141)
- PC1 of Sengkang site (172.17.0.140)
- PC1 of Tampines (172.19.0.130)

These test will verify if VPN Tunneling, ISAKMP Handshake, and Reachability is achieved and done properly

(Below are the displayed tests)

Bedok site tests:

Pinging from PC1 to Sengkang PC1

```
C:\>ping 172.17.0.140

Pinging 172.17.0.140 with 32 bytes of data:

Reply from 172.17.0.140: bytes=32 time=126ms TTL=124
Reply from 172.17.0.140: bytes=32 time=51ms TTL=124
Reply from 172.17.0.140: bytes=32 time=44ms TTL=124
Reply from 172.17.0.140: bytes=32 time=45ms TTL=124

Ping statistics for 172.17.0.140:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 44ms, Maximum = 126ms, Average = 66ms

C:\>ipconfig

FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::260:47FF:FE33:BD0D
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 10.0.0.141
    Subnet Mask . . . . .: 255.255.255.192
    Default Gateway . . . . .: ::
                                   10.0.0.65
```

Pinging from PC1 to Tampines PC1

```
C:\>ping 172.19.0.130

Pinging 172.19.0.130 with 32 bytes of data:

Reply from 172.19.0.130: bytes=32 time=78ms TTL=124
Reply from 172.19.0.130: bytes=32 time=49ms TTL=124
Reply from 172.19.0.130: bytes=32 time=58ms TTL=124
Reply from 172.19.0.130: bytes=32 time=45ms TTL=124

Ping statistics for 172.19.0.130:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 45ms, Maximum = 78ms, Average = 57ms

C:\>ipconfig

FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::260:47FF:FE33:BD0D
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 10.0.0.141
    Subnet Mask . . . . .: 255.255.255.192
    Default Gateway . . . . .: ::
                                   10.0.0.65
```

Tampines site tests:

Pinging from PC1 to Sengkang PC1

```
C:\>ping 172.17.0.140

Pinging 172.17.0.140 with 32 bytes of data:

Reply from 172.17.0.140: bytes=32 time=54ms TTL=124
Reply from 172.17.0.140: bytes=32 time=53ms TTL=124
Reply from 172.17.0.140: bytes=32 time=64ms TTL=124
Reply from 172.17.0.140: bytes=32 time=58ms TTL=124

Ping statistics for 172.17.0.140:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 53ms, Maximum = 64ms, Average = 57ms

C:\>ipconfig

FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::260:47FF:FE33:BD0D
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 172.19.0.130
    Subnet Mask . . . . .: 255.255.255.192
    Default Gateway . . . . .: ::
                                   172.19.0.129
```

Pinging from PC1 to Bedok PC1

```
C:\>ping 10.0.0.141

Pinging 10.0.0.141 with 32 bytes of data:

Reply from 10.0.0.141: bytes=32 time=89ms TTL=124
Reply from 10.0.0.141: bytes=32 time=85ms TTL=124
Reply from 10.0.0.141: bytes=32 time=72ms TTL=124
Reply from 10.0.0.141: bytes=32 time=126ms TTL=124

Ping statistics for 10.0.0.141:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 72ms, Maximum = 126ms, Average = 93ms

C:\>ipconfig

FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::260:47FF:FE33:BD0D
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 172.19.0.130
    Subnet Mask . . . . .: 255.255.255.192
    Default Gateway . . . . .: ::
                                   172.19.0.129
```

Sengkang site tests:

Pinging from PC1 to Tampines PC1

```
C:\>ping 172.19.0.130

Pinging 172.19.0.130 with 32 bytes of data:

Reply from 172.19.0.130: bytes=32 time=131ms TTL=124
Reply from 172.19.0.130: bytes=32 time=59ms TTL=124
Reply from 172.19.0.130: bytes=32 time=58ms TTL=124
Reply from 172.19.0.130: bytes=32 time=53ms TTL=124

Ping statistics for 172.19.0.130:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 53ms, Maximum = 131ms, Average = 75ms

C:\>ipconfig

FastEthernet0 Connection:(default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::260:47FF:FE33:BD0D
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 172.17.0.140
    Subnet Mask . . . . .: 255.255.255.192
    Default Gateway . . . . .: ::
                                172.17.0.129
```

Pinging from PC1 to Bedok PC1

```
C:\>ping 10.0.0.141

Pinging 10.0.0.141 with 32 bytes of data:

Reply from 10.0.0.141: bytes=32 time=94ms TTL=124
Reply from 10.0.0.141: bytes=32 time=89ms TTL=124
Reply from 10.0.0.141: bytes=32 time=82ms TTL=124
Reply from 10.0.0.141: bytes=32 time=84ms TTL=124

Ping statistics for 10.0.0.141:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 82ms, Maximum = 94ms, Average = 87ms

C:\>ipconfig

FastEthernet0 Connection:(default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::260:47FF:FE33:BD0D
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 172.17.0.140
    Subnet Mask . . . . .: 255.255.255.192
    Default Gateway . . . . .: ::
                                172.17.0.129
```

Proof of connections:

Sengkang Exitrouter Peer connections:

```
ExitRouter_SK#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst          src          state          conn-id slot status
99.99.99.5   99.99.99.1   QM_IDLE        1088    0 ACTIVE
99.99.99.9   99.99.99.1   QM_IDLE        1097    0 ACTIVE
```

Tampines Exitrouter Peer connections:

```
exitRouterBK#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst          src          state          conn-id slot status
99.99.99.1   99.99.99.5   QM_IDLE        1023    0 ACTIVE
99.99.99.9   99.99.99.5   QM_IDLE        1068    0 ACTIVE
```

Bedok Exitrouter Peer connections:

```
exitRouterBK#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst          src          state          conn-id slot status
99.99.99.1   99.99.99.5   QM_IDLE        1023    0 ACTIVE
99.99.99.9   99.99.99.5   QM_IDLE        1068    0 ACTIVE
```

4.2 Testing and Validation(Internet access verification)

Tampines site NSLOOKUP and PING test (To Internet Web/DNS Server)

```
C:\>nslookup sksite
Server: [200.100.50.254]
Address: 200.100.50.254

Non-authoritative answer:
Name: sksite
Address: 88.88.88.1

C:\>ping 200.100.50.254

Pinging 200.100.50.254 with 32 bytes of data:

Reply from 200.100.50.254: bytes=32 time=29ms TTL=124
Reply from 200.100.50.254: bytes=32 time=23ms TTL=124
Reply from 200.100.50.254: bytes=32 time=34ms TTL=124
Reply from 200.100.50.254: bytes=32 time=27ms TTL=124
```

(Below are the displayed configurations)

IP NAT translations

```
exitRouter_TMP#show ip nat translation
Pro Inside global      Inside local      Outside local      Outside global
icmp 66.220.149.1:31    172.19.0.130:31   200.100.50.254:31  200.100.50.254:31
icmp 66.220.149.1:32    172.19.0.130:32   200.100.50.254:32  200.100.50.254:32
icmp 66.220.149.1:33    172.19.0.130:33   200.100.50.254:33  200.100.50.254:33
icmp 66.220.149.1:34    172.19.0.130:34   200.100.50.254:34  200.100.50.254:34
udp  66.220.149.1:1028  172.19.0.130:1028 200.100.50.254:53  200.100.50.254:53
```

(Udp and ICMP protocols translated from PC1)

Bedok site NSLOOKUP and PING test (To Internet Web/DNS Server)

```
C:\>nslookup bedoksite

Server: [200.100.50.254]
Address: 200.100.50.254

Non-authoritative answer:
Name:   bedoksite
Address: 88.88.88.5

C:\>ping 200.100.50.254

Pinging 200.100.50.254 with 32 bytes of data:

Reply from 200.100.50.254: bytes=32 time=27ms TTL=124
Reply from 200.100.50.254: bytes=32 time=85ms TTL=124
Reply from 200.100.50.254: bytes=32 time=38ms TTL=124
Reply from 200.100.50.254: bytes=32 time=49ms TTL=124

Ping statistics for 200.100.50.254:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 27ms, Maximum = 85ms, Average = 49ms
```

IP NAT translations

```
exitRouterBK>en
exitRouterBK#show ip nat translations
Pro Inside global      Inside local      Outside local      Outside global
icmp 154.26.131.1:17    10.0.0.141:17     200.100.50.254:17  200.100.50.254:17
icmp 154.26.131.1:18    10.0.0.141:18     200.100.50.254:18  200.100.50.254:18
icmp 154.26.131.1:19    10.0.0.141:19     200.100.50.254:19  200.100.50.254:19
icmp 154.26.131.1:20    10.0.0.141:20     200.100.50.254:20  200.100.50.254:20
icmp 154.26.131.1:21    10.0.0.141:21     200.100.50.254:21  200.100.50.254:21
icmp 154.26.131.1:22    10.0.0.141:22     200.100.50.254:22  200.100.50.254:22
icmp 154.26.131.1:23    10.0.0.141:23     200.100.50.254:23  200.100.50.254:23
icmp 154.26.131.1:24    10.0.0.141:24     200.100.50.254:24  200.100.50.254:24
udp  154.26.131.1:1025  10.0.0.141:1025   200.100.50.254:53  200.100.50.254:53
udp  154.26.131.1:1027  10.0.0.139:1027   200.100.50.254:53  200.100.50.254:53
```

(Udp and ICMP protocols translated from PC1)

Sengkang site NSLOOKUP and PING test (To Internet Web/DNS Server)

```
C:\>nslookup bedoksite
Server: [200.100.50.254]
Address: 200.100.50.254

Non-authoritative answer:
Name: bedoksite
Address: 88.88.88.5

C:\>ping 200.100.50.254

Pinging 200.100.50.254 with 32 bytes of data:

Reply from 200.100.50.254: bytes=32 time=49ms TTL=124
Reply from 200.100.50.254: bytes=32 time=29ms TTL=124
Reply from 200.100.50.254: bytes=32 time=27ms TTL=124
Reply from 200.100.50.254: bytes=32 time=26ms TTL=124

Ping statistics for 200.100.50.254:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 26ms, Maximum = 49ms, Average = 32ms
```

IP NAT translations

```
ExitRouter_SK#show ip nat translations
Pro Inside global      Inside local      Outside local      Outside global
icmp 64.233.185.113:15 172.17.0.140:15   200.100.50.254:15 200.100.50.254:15
icmp 64.233.185.113:16 172.17.0.140:16   200.100.50.254:16 200.100.50.254:16
icmp 64.233.185.113:17 172.17.0.140:17   200.100.50.254:17 200.100.50.254:17
icmp 64.233.185.113:18 172.17.0.140:18   200.100.50.254:18 200.100.50.254:18
udp  64.233.185.113:1027 172.17.0.140:1027 200.100.50.254:53 200.100.50.254:53
```

(Udp and ICMP protocols translated from PC1)

4.3 Testing and Validation(Firewall functionality)

Part A) testing specifically only allowing PC1 to SSH into ASA

Sengkang SSH into ASA from PC1 (allowed)

```
C:\>ssh -l Admin01 172.17.0.129

Password:

Site-ASA5506>
```

Sengkang SSH into ASA from PC0 (not allowed)

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ssh -l Admin01 172.17.0.129

% Connection timed out; remote host not responding
C:\>|
```

Tampines SSH into ASA from PC1 (allowed)

```
C:\>ssh -l Admin01 172.19.0.129
Password:
Site-ASA5506>
```

Tampines SSH into ASA from PC0 (not allowed)

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ssh -l Admin01 172.19.0.129

% Connection timed out; remote host not responding
C:\>
```

Bedok SSH into ASA from PC1 (allowed)

```
C:\>ssh -l Admin01 10.0.0.129
Password:
SITE-ASA5506>
```

Bedok SSH into ASA from PC0 (not allowed)

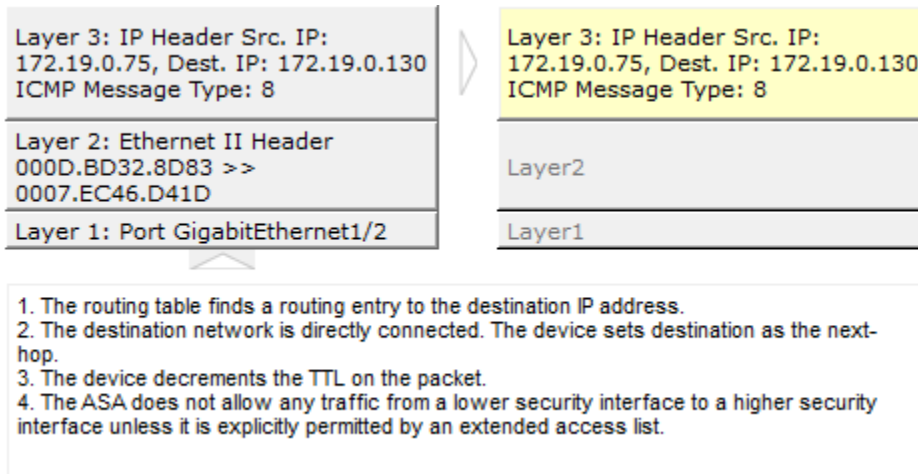
```
C:\>ssh -l Admin01 10.0.0.129

% Connection timed out; remote host not responding
C:\>
```

Part B) Attempting to ping from a zone with lower security level to a zone with higher security levels:

All 3 used interfaces of the ASA are configured with proper security levels, hence this is the output of the ping test

(Below are the displayed configurations)



```
Cisco Packet Tracer SERVER Command Line 1.0
C:\>ping 172.19.0.130

Pinging 172.19.0.130 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 172.19.0.130:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Attempt to ping from dmz server to LAN PC

Part C) Stateful Inspection

Policy-Map on ASA

```
Site-ASA5506#show run | section policy
policy-map type inspect dns preset_dns_map
  parameters
    message-length maximum 512
policy-map global_policy
  class inspection_default
    inspect dns preset_dns_map
    inspect ftp
    inspect icmp
    inspect tftp
service-policy global_policy global
```

- Inspect ICMP, FTP, TFTP, DNS
- This allows replies even though the packet came from a zone of a lower security interface

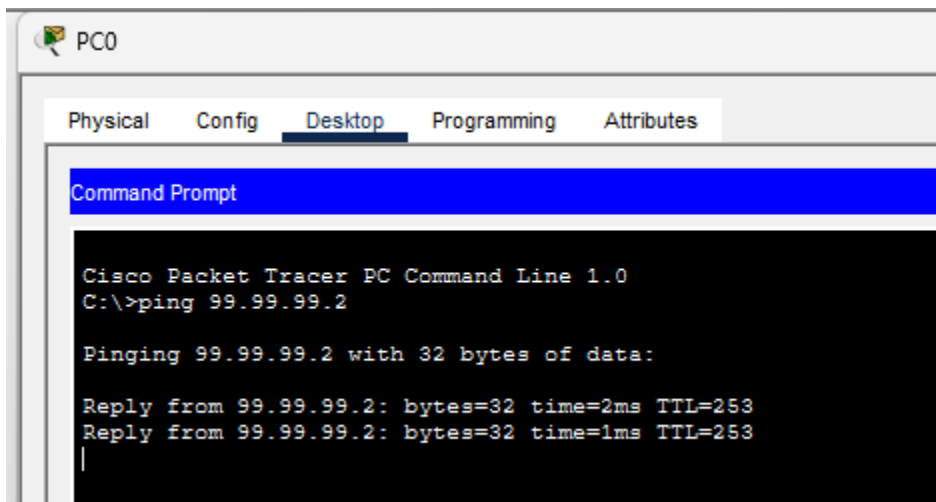
As testing, this is the only access-list applied on the outside interface

```
access-list OUTSIDE-IN extended permit ip 172.19.0.0 255.255.224.0 172.17.0.0
255.255.128.0
access-list OUTSIDE-IN extended permit ip 10.0.0.0 255.128.0.0 172.17.0.0 255.255.128.0
!
!
access-group OUTSIDE-IN in interface outside
```

It does not have any protocols to permit packets into ASA from internet public IPs, thus by default should drop packets

The ASA by default due to security level differences, should drop the packet too

This test will attempt to ping from the PC to the OUTSIDE



As you can see, echo replies is still permitted to the PCs, even though it is in the LAN.

The next test will attempt to ping the same, but without inspect ICMP

```
policy-map type inspect dns preset_dns_map
  parameters
    message-length maximum 512
policy-map global_policy
  class inspection_default
    inspect dns preset_dns_map
    inspect ftp
    inspect tftp
```

(no inspect icmp)

(Below are the displayed configurations)

Result: Ping never replies back

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 99.99.99.1

Pinging 99.99.99.1 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 99.99.99.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

This shows the stateful inspection is working as intended, it allows tracking of packet state to reply to packet origin

Part D) ACLs for Site-to-Site connectivity

Normally, pinging site to site will fail because the ASA in the receiving set will treat that as OUTSIDE -> INSIDE, which is not allowed because of security zones.

However, as mentioned earlier, with ACLs, the ASA can allow this and do it safely as it is across a **private VPN Tunnel**.

Access-list implemented on the outside

Bedok

```
SITE-ASA5506#show run | include OUTSIDE-IN
access-list OUTSIDE-IN extended permit ip 172.19.0.0 255.255.224.0 10.0.0.0 255.128.0.0
access-list OUTSIDE-IN extended permit ip 172.17.0.0 255.255.192.0 10.0.0.0 255.128.0.0
access-group OUTSIDE-IN in interface outside
-----
```

Tampines

```
Site-ASA5506#show run | include OUTSIDE-IN
access-list OUTSIDE-IN extended permit ip 172.17.0.0 255.255.128.0 172.19.0.0
255.255.224.0
access-list OUTSIDE-IN extended permit ip 10.0.0.0 255.128.0.0 172.19.0.0 255.255.224.0
access-group OUTSIDE-IN in interface OUTSIDE
-----
```

Sengkang

```
access-list OUTSIDE-IN extended permit ip 172.19.0.0 255.255.224.0 172.17.0.0
255.255.128.0
access-list OUTSIDE-IN extended permit ip 10.0.0.0 255.128.0.0 172.17.0.0 255.255.128.0
!
!
access-group OUTSIDE-IN in interface outside
```

Successful results of Site-To-Site Pinging was shown in section 4.1

5.1 Miscellaneous Configurations(SSh)

SSH was configured on each network device with a different SSH credential from ASA SSH credentials

Other router/switches credentials

```
!  
username netAdmin privilege 15 secret 5 $1$mERr$Pf9qnibOLaOxNtZyA4tEf0  
!
```

Password: iLOVEnws

Example of Sengkang PC1 SSH into routers/switches successfully

```
C:\>ssh -l netAdmin 172.17.0.130  
  
Password:  
LAN-Switch>  
LAN-Switch>exit  
  
[Connection to 172.17.0.130 closed by foreign host]  
C:\>ssh -l netAdmin 99.99.99.2  
  
Password:  
  
ISP_Edge_SK#
```

Example of Bedok PC1 SSH into routers/switches successfully

```
Cisco Packet Tracer PC Command Line 1.0  
C:\>ssh -l netAdmin 10.0.0.80  
  
Password:  
  
DMZswitch>exit  
  
[Connection to 10.0.0.80 closed by foreign host]  
C:\>ssh -l netAdmin 99.99.99.5  
  
Password:  
  
exitRouterBK#
```

Example of Tampines PC1 SSH into routers/switches successfully

```
Cisco Packet Tracer PC Command Line 1.0  
C:\>ssh -l netAdmin 172.19.0.135  
  
Password:  
  
WARNING: AUTHORIZED LAN ACCESS ONLY  
TMP-SW1>exit  
  
[Connection to 172.19.0.135 closed by foreign host]  
C:\>ssh -l netAdmin 99.99.99.10  
  
Password:  
% Login invalid  
  
Password:  
  
WARNING: NWS ISP PROPERTY - AUTHORIZED ACCESS ONLY  
ISP_Edge_TP#
```

Network devices (Switch/Routers) were also configured with the basic SSH configurations:

```
line vty 0 4
  exec-timeout 20 0
  login local
  transport input ssh
line vty 5 15
  exec-timeout 20 0
  login local
  transport input ssh
```

ip domain-name NWS.com

username netAdmin privilege 15 secret iLOVEnws

crypto key generate rsa

1024

line vty 0 15

transport input ssh

login local

6. Challenges Faced

Describe any technical or collaborative challenges encountered during the project and how they were resolved.

NTP

```
Site-ASA5506#show ntp associations

address      ref clock    st  when    poll    reach  delay    offset
disp
~172.17.0.5   .AUTH.       16  -        64      0      0.00     0.00
0.01
+-----+-----+-----+-----+-----+-----+-----+-----+
SITE-ASA5506(config)#show ntp status
Clock is unsynchronized, stratum 16, no reference clock
nominal freq is 250.0000 Hz, actual freq is 249.9990 Hz, precision is 2**24
reference time is 00000000.00000000 (00:00:00.000 UTC Mon Jan 1 1990)
clock offset is 0.00 msec, root delay is 0.00 msec
root dispersion is 0.00 msec, peer dispersion is 0.00 msec.
loopfilter state is 'FSET' (Drift set from file), drift is - 0.000001193 s/s system poll interval
is 4, never updated.
SITE-ASA5506(config)#
```

One of the technical challenges that we encountered during the project was the failure of the Cisco ASA 5506 firewall to synchronise its clock using NTP. Despite configuring an internal NTP server within the Server Farm, ASA consistently reported an unsynchronised clock.

To fix this, we used simulation mode to find out where the packet for NTP was being dropped and the reason why. Otherwise, when simulation mode carries out the NTP authentication exchange successfully. We temporarily added an access-list to allow NTP to synchronise with the clock. Once it was successful, we removed the NTP access list to ensure that it would not be used for any malicious activity

VPN

VPN ISAKMP handshakes were not able to be established across sites

This could be seen running *show crypto isakmp sa*

The issue was that although VPN peers IP were configured correctly, with matching access-lists, the static routes were missing on the ISP router, preventing the negotiation packets from being routed to the destination peer.

This cause *show crypto isakmp sa* to display MM_NO_STATE

To fix this, we added static routes in ISP router to route 99.0.0.0/8 (according to the subnet) to the exit routers, so VPN Tunnels can be established

Bedok

```
ISProuter#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    99.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       99.99.99.4/30 is directly connected, Serial0/2/1
L       99.99.99.6/32 is directly connected, Serial0/2/1
       154.26.0.0/24 is subnetted, 1 subnets
S       154.26.131.0/24 [1/0] via 99.99.99.5
```

Tampines

```
ISP_Edge_IP#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    66.0.0.0/25 is subnetted, 1 subnets
S       66.220.149.0/25 [1/0] via 99.99.99.9
    99.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       99.99.99.8/30 is directly connected, Serial0/2/1
L       99.99.99.10/32 is directly connected, Serial0/2/1
```

Sengkang

```
ISP_Edge_SK#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 99.99.99.1 to network 0.0.0.0

    64.0.0.0/28 is subnetted, 1 subnets
S       64.233.185.112/28 [1/0] via 99.99.99.1
    99.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       99.99.99.0/30 is directly connected, Serial0/2/1
L       99.99.99.2/32 is directly connected, Serial0/2/1
S*     0.0.0.0/0 [1/0] via 99.99.99.1
```

This allows ISAKMP handshakes to be completed and for the phase 1 of the VPN tunnel to display correct connections:

Sengkang Exitrouter Peer connections:

```
ExitRouter_SK#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst          src          state          conn-id slot status
99.99.99.5    99.99.99.1    QM_IDLE       1088      0 ACTIVE
99.99.99.9    99.99.99.1    QM_IDLE       1097      0 ACTIVE
```

Tampines Exitrouter Peer connections:

```
exitRouterBK#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst          src          state          conn-id slot status
99.99.99.1    99.99.99.5    QM_IDLE       1023      0 ACTIVE
99.99.99.9    99.99.99.5    QM_IDLE       1068      0 ACTIVE
```

Bedok Exitrouter Peer connections:

```
exitRouterBK#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst          src          state          conn-id slot status
99.99.99.1    99.99.99.5    QM_IDLE       1023      0 ACTIVE
99.99.99.9    99.99.99.5    QM_IDLE       1068      0 ACTIVE
```

SSH

When doing SSH into network devices, even after assigning an IP to the VLAN in layer 2 switches, the PC is unable to connect to the layer 2 switch. This is because layer 2 switches cannot route like layer 3.

An example of the error faced

```
C:\>ssh -l netAdmin 172.17.0.66
% Connection timed out; remote host not responding
C:\>D
```

The fix to this was adding a ip default-gateway to the layer 2 switches allowed it to know where to send the TCP packet, allowing SSH connections

Added default gateway to layer 2 switches, to route towards the ASA Gig1/2 interface

Example on Sengkang Site DMZ Switch

```
!
interface Vlan1
 ip address 172.17.0.66 255.255.255.192
!
ip default-gateway 172.17.0.65
!
```

Once that is configured, the layer 2 switch knows where to send back the TCP Connection now

```
C:\>ssh -l netAdmin 172.17.0.66  
Password:  
Server-switch>
```

Able to SSH into layer 2 switches now

(This part is intentionally left blank)

Part B – Zero Trust Research

Overview

The Zero Trust Architecture operates under a “never trust, always verify”. Unlike traditional security models, Zero Trust believes that threats can exist not only outside the network but also inside.

Every access request is treated as if it originates from an open network, making sure all requests are verified through strict identity authentication, device validation and real-time policy checks. It follows the principle of least privilege, granting only the minimum necessary access. Even after access is granted, Zero Trust continuously monitors user behavior and device status, so if there is suspicious activity, it can be detected and blocked immediately. This prevents unauthorized lateral movement within the network and helps secure sensitive resources from internal and external threats.



Principles of zero trust

Why it matters for this network

Our network consists of multiple geographically distributed sites each with its own internal LAN, server farm, firewall and exit router. The sites are interconnected through the public using site-to-site VPN tunnels, with NAT configured at each exit router to allow controlled internet access. Within each site, network segmentation is enforced through separate zones.

Given this topology, if any device were to be compromised within the network, it poses a significant risk. An attacker who gains access to one LAN device or server could attempt lateral

movement between the segments of our network and even use the VPN tunnels to reach other sites.

Implementing Zero Trust Principles in our network removes the assumption that all traffic is trusted. By enforcing strict access controls on each device, zero trust can help contain potential breaches, reduce lateral movement and ensure that a compromise at one site does not impact the rest of the network.

What can be changed in our network to implement Zero Trust

1. Overview of the Existing Network Architecture

As mentioned above, each site has a similar structure, comprising an internal LAN, server farm, firewall and an exit router that connects the site to the internet and Inter-site communication is achieved through a mesh of site-to-site VPN tunnels.

Security controls that already exist include NAT at the exit router, firewall zoning using INSIDE, OUTSIDE and DMZ interfaces on the ASA firewall and DHCP services for the internal hosts, and access control policies to regulate traffic. These controls provide a strong security foundation that align with network security best practices.

However, it is still built on a trusted internal network. Once a user gains access to any device within any site, they are able to perform attacks like lateral movement to gain access to more devices since the traffic is generally treated as trusted.

2. Strengthening Segmentation with Zero Trust Principles

A core principle of Zero trust is minimising implicit trust within the network, even for devices that are already within the internal traffic. The current architecture already separates traffic using VLANs and ASA security zones, Zero trust requires segmentation to be more granular and policy driven rather than trusting all internal traffic.

Under the proposed Zero Trust design, segmentation would be enforced not only at the perimeter but also within each site. Internal LANs, server farms and management networks should be treated as independent trust zones, with explicit access rules governing communication between them. This reduces the chances that a compromised internal host can freely access sensitive servers or infrastructure components

How can this be achieved?

Strengthening Segmentation

This can be done by only allowing user LAN devices to be allowed to only access the necessary services that they require on the designated servers. Direct access from user LANs to server management interfaces should be denied by default. This also applies to server-to-server communication if it were to be implemented in the future where it should be restricted to what is only required

The next thing that can be implemented is applying least privilege access at the ASA firewall. The ASA firewall policies should be changed from being permissive, network-wide rules to least-privilege and more application-aware access controls.

Applying least privilege to firewall

This can be done by applying access rules that are defined based on the source role, user LAN, server farm, etc., destination service, protocol and direction of traffic. For example, rather than allowing all INSIDE traffic to the DMZ, the ASA should only permit from USER LAN to Web Server and DNS server. As well as Admin PC to server farm only allowing management ports like SSH. All other unnecessary traffic should be denied by default and logged. This ensures that if a device is compromised, they cannot access unnecessary services or infrastructure components .

Site-to-site VPN connections

In the current design, the site-to-site VPNs only provide encryption to protect data in transit. However, Zero Trust requires that encryption does not simply imply trust. (reference)

To fix this, VPN traffic filtering and firewall policies to enforce inter-site communications are restricted. Each site should only be allowed to access specific devices at other sites and specific services required for operations.

For example, a user LAN in the Bedok site should not be able to access the LAN devices in the Tampines site unless it is required. Instead, access should be limited to shared services such as servers or applications.

By implementing Zero Trust between sites, the risk that the other sites get compromised due to one compromised site is greatly reduced.

3. Identity, Authentication & Continuous Verification

Shifting from Network-Based Trust to Identity-Based Access

In the current NWS network, access decisions are based on network location, while this approach is effective for perimeter security, it does not align fully with Zero trust principles which emphasize identity as the security perimeter.

To adopt Zero Trust, access control should be shifted from where the traffic comes from to who or what is requesting access. This includes user identity, device identity and service identity. Under this model, even if a device is connected to the internal LAN or through a VPN tunnel, access is not automatically trusted without validating identity and authorisation.

This can be introduced by enforcing stronger authentication methods for users, administrators and network devices before allowing access to internal services or management interfaces.

Strengthening Authentication for Users and Administrators

Zero trust is heavily dependent on the use of strong authentication. Currently, administrative access to network devices such as the ASA firewall relies on local AAA accounts and SSH restrictions. While this provides basic security, Zero Trust recommends stronger and more consistent authentication enforcement.

Improvements that can be done include centralising authentication using AAA services for administrators. Enforcing strong credentials and role-based access for different administrative tasks. Restricting administrative access to specific devices and services based on role.

For user access to internal services, authentication should be required even for internal applications. This ensures that a compromised device once again does not automatically gain access to sensitive resources.

Visibility, Logging, and Policy Enforcement

Visibility makes it easier for Zero trust to make informed access decisions. Proper logging and monitoring as well as continuous verification makes this possible. Therefore, logging of events such as authentication attempts (for both successful and failed), inter-segment traffic flows and VPN-based access between sites should be recorded.

These provide valuable insights into normal and abnormal behaviour, enabling faster detection of potential threats. In a real-world deployment, this information would feed into a central monitoring system of SIEM for correlation and analysis.

Continuous Verification and Session-Based Access Control

In our current network, access is verified only once, at the point of entry such as during an initial login or VPN connection. Once authenticated, users and devices are often allowed to maintain access for extended periods with minimal revalidation. This approach increases the risk, as attackers who successfully compromise credentials or sessions may retain long-lived access without detection.

Under a Zero Trust architecture, access is treated as temporary and conditional, requiring continuous verification throughout the duration of a session. For the NWS network, this principle can be applied by introducing stricter session controls across administrative access, inter-site communication and internal services.

Administrative access to critical infrastructure such as the ASA firewall, routers and servers should be subject to shorter session timeouts and enforced re-authentication. For example, SSH sessions that remain idle beyond a defined threshold should be automatically terminated, reducing the risk of unattended or hijacked sessions. This ensures that access remains active only when it is actively being used by an authorised administrator.

This approach aligns with the Zero Trust assumption that breaches are possible and focuses on limiting attacker dwell time, improving detection and maintaining tighter control over access throughout the network lifecycle.

Conclusion

The existing NWS network offers a good starting point with the incorporation of firewalls, VPNs, and network segmentation. Despite that, with the increasing level of sophistication and distribution of threats, relying on perimeter security is no longer adequate. By implementing a Zero Trust architecture, the network can move away from the traditional trust-by-location approach and focus on real-time enforcement of access based on policy, context, and need.

This proposal illustrates the practical implementation of Zero Trust on the existing multi-site Packet Tracer network topology. By doing so, the NWS network will be rendered more resilient, secure, and better equipped to counter contemporary cyber threats.

6. Conclusion

This project successfully demonstrated the design, implementation, and evaluation of a secure multi-site enterprise network using Cisco Packet Tracer, while exploring how Zero Trust principles can be applied to strengthen its security posture. During Part A, a full-mesh site-to-site VPN architecture was implemented across the Sengkang, Tampines, and Bedok sites, supported by NAT, Cisco ASA firewalls, and structured network segmentation. These configurations ensured secure inter-site connectivity, controlled internet access, and effective perimeter protection, meeting the technical objectives of the Proof of Concept.

While the existing design provides a strong foundation through encryption, firewall zoning, and access control, Part B highlighted that traditional perimeter-based security alone is insufficient against modern threats. The Zero Trust research component identified key trust assumptions within the network, particularly the implicit trust placed on internal traffic and VPN-connected sites. By proposing enhancements such as granular segmentation, least-privilege firewall policies, identity-based access control, continuous verification, and improved monitoring, the network can better contain breaches, reduce lateral movement, and limit attacker dwelling time.

Importantly, this proposal demonstrates that Zero Trust does not require a complete replacement of the existing architecture. Instead, it can be incrementally integrated into the current network by refining policies, strengthening authentication, and enhancing visibility across sites. This makes the Zero Trust approach both technically feasible and realistic within a proof-of-concept environment.

Overall, the project illustrates how traditional enterprise networking concepts and modern Zero Trust principles can work together to create a more resilient and secure infrastructure. By combining practical configuration skills with forward-looking security design, the NWS network is better positioned to defend against contemporary cyber threats while maintaining scalability, operational efficiency, and industry standard security.

7. References

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